

Experimental investigation on rockburst process and failure characteristics in trapezoidal tunnel under different lateral stresses

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HIGHLIGHTS

- High lateral stress accelerates the formation of localized damage around the trapezoidal opening.
- High lateral stress can restrain the rotation of the principal stress.
- Supporting area of trapezoidal opening is larger than that of circular opening.
- Damage degree at the sidewall of opening increases with increasing lateral stress.

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ABSTRACT

In general, the in-situ stress increases with the increase of excavation depth, and it significantly affects the safe and long-term stability of tunnels and chambers. To study the influence of the in-situ stress on the rockburst process and failure characteristics, three different lateral stresses were loaded to perform biaxial compression experiments on sandstone specimens with a trapezoidal opening. The results indicate that the high lateral stress accelerates the formation of localized damage around the trapezoidal opening, which can effectively consume the elastic strain energy. The deep surrounding rock at the roof and floor are basically in a compressive stress state, and the tangential compressive stress in the deep surrounding rock of the sidewall is greater than that in the deep surrounding rock of the roof and floor, which can cause the deep surrounding rock of the sidewall to be damaged more severely. For the corners of the trapezoidal opening, the high lateral stress can restrain the rotation of the principal stress, which lead to the reduction of the crack density around the corners. AE signals and CT scanning show that the failure mode of the model specimen shifts from tensile fracture to mixed tensile and shear fracture with the increase of the lateral stress. Under different lateral stresses, the outlines of failure zones on both sidewalls of trapezoidal opening are V-shaped, and the damage degree at the sidewall of the trapezoidal opening increases with the increase of the lateral stress when the rockburst occurs. Additionally, the failure zones on both sidewalls are larger than those of circular opening, which indicate that the supporting area on the sidewall of trapezoidal opening needs to be larger than that on the sidewall of circular opening.

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1. Introduction

To satisfy the requirements of coal and metal mining, transportation engineering construction, energy storage, and high-level radioactive waste disposal, deep underground rock engineering has emerged and rapidly develops. These rock engineering constructions generally contain cavities or tunnels excavations of rock

masses in deep underground areas, which involve more and more complex rock mechanics problems [1]. Deep excavation causes high stress concentration around the opening, which is the main reason for fracturing in the side wall slab as well as rockburst in the most practical cases encountered in mining and tunneling [2,3]. Such damages can result in economic loss and human casualties. Thus, the stability of underground openings in deep rock masses has become a huge hidden safety hazard in the process of cavity or tunnel excavation, which has severely hindered the smooth development of underground rock engineering construc-

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tion [4]. As a result, deep understandings of the fracture behavior around openings and the development process and mechanism of rockburst are crucial for eliminating rockburst hazard and thus improving the stability of cavities or tunnels.

To investigate rockburst, numerous researches have been conducted in deep rock, and considerable results have been obtained [5–13]. Wang and Park [10] investigated the property of strain energy stored in rocks and the study showed that strain energy analysis with the aid of numerical modeling was useful in prediction of rockburst. He et al. [11] conducted experiments under different stress paths and experimental methods, and classified the rockburst into two major groups, i.e., the strainburst and impact-induced burst. Mazaira and Konicek [12] summed up the influencing factors of rockburst involving the geological and geomechanical factors and the engineering measures taken to control rockburst.

Numbers deep engineering cases show that rockburst often occurs on the sidewall of cavern, and V-shaped notches are formed at both sidewalls [14,15]. For the laboratory simulation of this phenomenon, some scholars have deeply studied this idea. Using a sandstone specimen with a circular opening and particle flow code simulation software, Fakhimi et al. [16] simulated the failure process around an underground excavation and modeled the observed damage zone around the opening. Biaxial compression experiments were conducted by Zhang et al. [17] to investigate the rockburst process of a granite tunnel with a circular section under horizontal stress conditions. Hu et al. [18] made the structural model using granite specimens with a circular hole and investigated the rockburst process and found that a sharp increase in AE energy can serve as an early warning signal for overall rockburst. In addition, they determined that rockburst can be predicted by considering AE event density as a potential indicator. Red sandstone samples with a circular hole were tested under four different initial stress conditions [19]. The experimental results showed that the effect of lateral stress on spalling damage and rockburst was more obvious than that of axial stress. The aforementioned studies mainly focused on the rockburst in circular caverns. However, researchers have determined that the opening shape has a significant influence on the failure characteristics and rockburst process [20–26]. Zhu et al. [20] simulated progressive fracturing processes based on three shapes generally used in practical underground engineering: circular, elliptical, and inverted U-shapes. Their study analyzed the stress distribution in rock around different opening shapes and the resulting fracture patterns. Li et al. [23] applied digital image correlation in separate investigations of the fracturing process of marble containing a circular and an elliptical opening. Experimental results showed that the nucleation and propagation of strain localization zones dominated the propagation of tensile cracks surrounding the opening. Zhong et al. [24] investigated the initiation, propagation, and coalescence of cracks around an inverted U-shaped opening under the biaxial compression condition and found that rather than splitting and rockburst, slight fracture-induced damage occurred, particularly shear slipping in the sidewall. Luo et al. [25] studied the process and failure characteristics of sidewall spalling in D-shaped tunnels. The results showed that failure zone on both sidewalls were located between the corner and spandrel and the D-shaped tunnels could effectively reduce the intensity of rock failure and transform the rockburst into a static spalling compared with the circular tunnels. However, few reports have focused on the fracture behavior and rockburst process around a trapezoidal opening, which is a common shape used in underground excavation. In addition, with the increase of the excavation depth or the influence of geologic structure, the horizontal lateral stress would change, which would affect the failure characteristics of caverns. Therefore, it is necessary to conduct such researches on these topics.

In this study, biaxial compressive experiments with different lateral stresses were conducted on actual rock specimens containing trapezoidal openings. In addition, applications of a high-speed camera, strain measurement system, and AE monitoring system enabled us to obtain detailed analyses of strain distribution, fracture propagation, fracture patterns, and the rockburst.

2. Experimental apparatus and scheme

2.1. Specimen description and preparation

The sandstone used in the present study was sourced from an outcrop in the Chayuan New Area east of the South Bank of the Yangtze River in Chongqing, China. The uniaxial compressive strength (UCS) and the tensile strength of the sandstone were 46.90 MPa and 3.3 MPa, respectively. Three rectangular prism specimens were taken from a single block of sandstone, and their dimensions were 100 mm × 100 mm × 50 mm in width, height, thickness. After cutting, the specimens were polished carefully to ensure that the relative non-parallel error between the faces of opposite ends did not exceed 0.02 mm. Then, a trapezoidal hole was precisely machined in the center of the specimen along the thickness direction by using a high-pressure water-jet cutting technique. The dimensions of the trapezoidal hole were 15 mm × 10 mm × 15 mm for the floor, roof, height as shown in Fig. 1a. During the preparation of the specimen, silicone rubber was applied to prevent breakage in the wiring of the strain rosette gauges during the analysis. In addition, because the cracking process in the front of the specimen needed to be monitored and recorded in real time during the experiment, two AE sensors were placed near the back opening of the specimen, as shown in Fig. 1b. In order to comprehensively measure the deformation around the opening, three model 120-3AA strain resistance gauges with a 7-mm-long measuring grid (R3, S6, F9) were bonded to the inside of the opening, and six resistance strain gauges in total were bonded to the surface of the specimen at the roof (R1, R2), sidewall (S4, S5), and floor (F7, F8). Two model 120-3CA three-dimensional strain rosette gauges were bonded to the upper corner (C10) and lower corner (C11), respectively. The spacing between the strain rosette gauges and the opening boundary were 5 mm and 15 mm, respectively (Fig. 1c).

2.2. Experimental apparatus and procedure

The entire experimental system consisted of a loading apparatus, an AE system, deformation acquisition equipment, and an integrated observation system, as shown in Fig. 2. The uniaxial and biaxial compression experiments in this study were performed by using the true triaxial geophysical apparatus (TTG) independently developed by Chongqing University [27]. The apparatus was used for compression experiments of uniaxial, biaxial, and true triaxial stresses. In addition, it can apply maximum three-dimensional (3D) stress levels of 6000 kN in two directions and 4000 kN in a different direction. Deformation of the specimen in the three directions was measured by using six linear variable differential transformers (LVDTs). The PAC PCI-2 AE monitoring system was applied to record the AE characteristics of the specimens during the entire experiment process. A strain meter was utilized to measure the deformation behaviors of the specimens. The integrated observation system, which includes a high-speed camera and digital camera, was used to acquire the visible images that were used to conduct inversion analysis of the evolution process of the crack propagation. After the biaxial compression experiments were conducted, all the specimens were scanned by

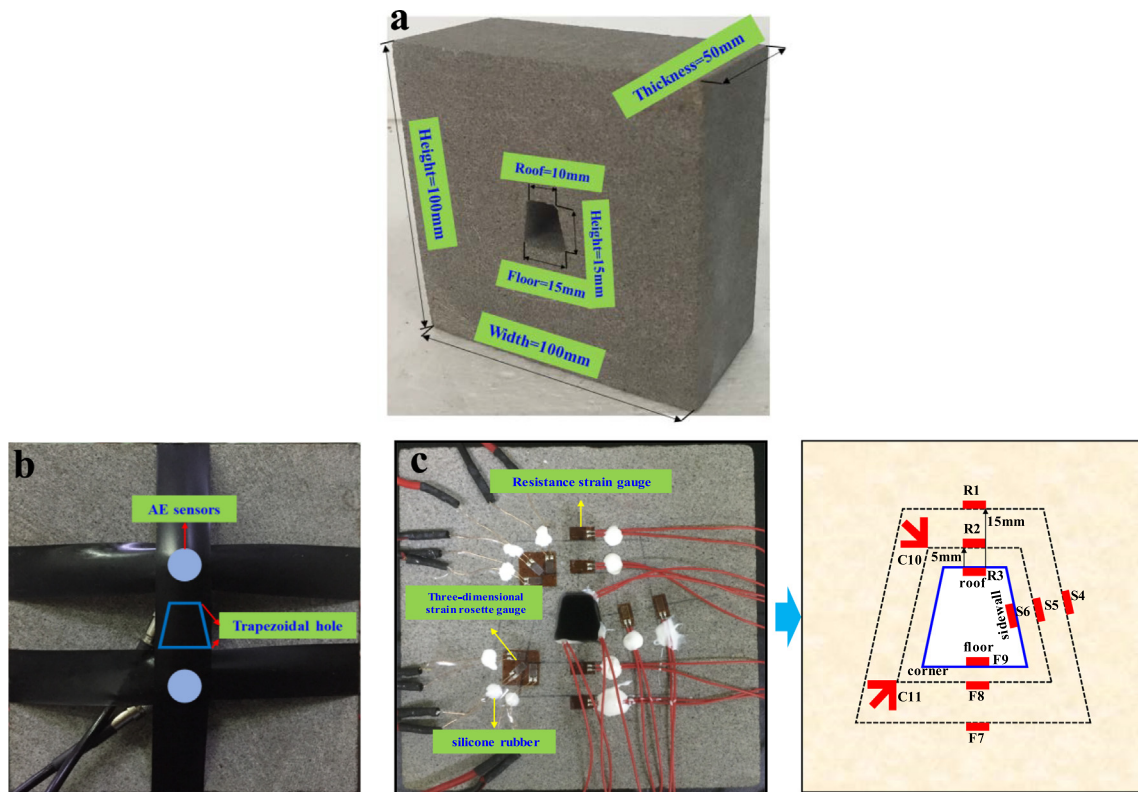


Fig. 1. Experimental specimen: (a) specimen size; (b) layout for AE transducers; (c) layout for strain gauges.

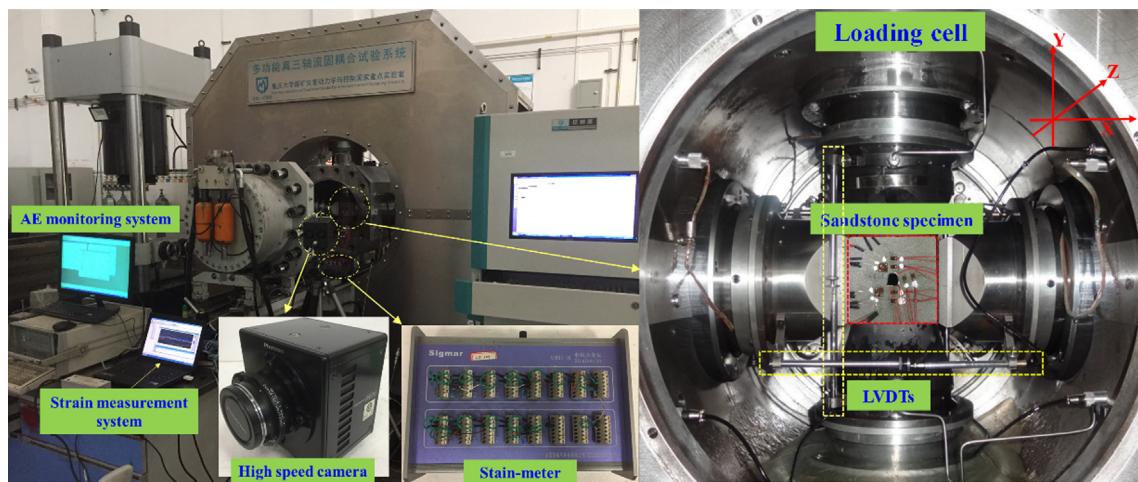


Fig. 2. Experimental system containing loading apparatus, acoustic emission system, deformation acquisition equipment, and integrated observation system.

X-ray computed tomography (CT) to identify the fracture patterns within the opening.

The rock mass is under a true-triaxial stress state before excavation. With the advancement of excavation, the tangential stress σ_θ of rock mass continues to build up until plane strain conditions are approximately formed [18]. As σ_θ exceeds the strength of rock mass, the residual strain energy after fracturing would be released in the form of kinetic energy, leading to the occurrence of intense rockburst. This phenomenon can be simulated in the laboratory by gradually increasing the vertical stress σ_v with a given lateral stress σ_h ,

which can form a relatively large deviatoric stress. In addition, the applied vertical stress σ_v and lateral stress σ_h on the structural sandstone specimen in this study are used to simulate the gradually increase of σ_θ .

According to the above analysis, we designed the loading path to approximately simulate the stress path of the rock mass during the excavation (Fig. 3).

Firstly, σ_v and σ_h were loaded to the preset magnitude. Next, the σ_h remained constant, while the σ_v was increased at a rate of 0.002 mm/s until violent rockburst occurred. The σ_h was between 0 and 20 MPa, i.e. 0, 10, and 20 MPa.

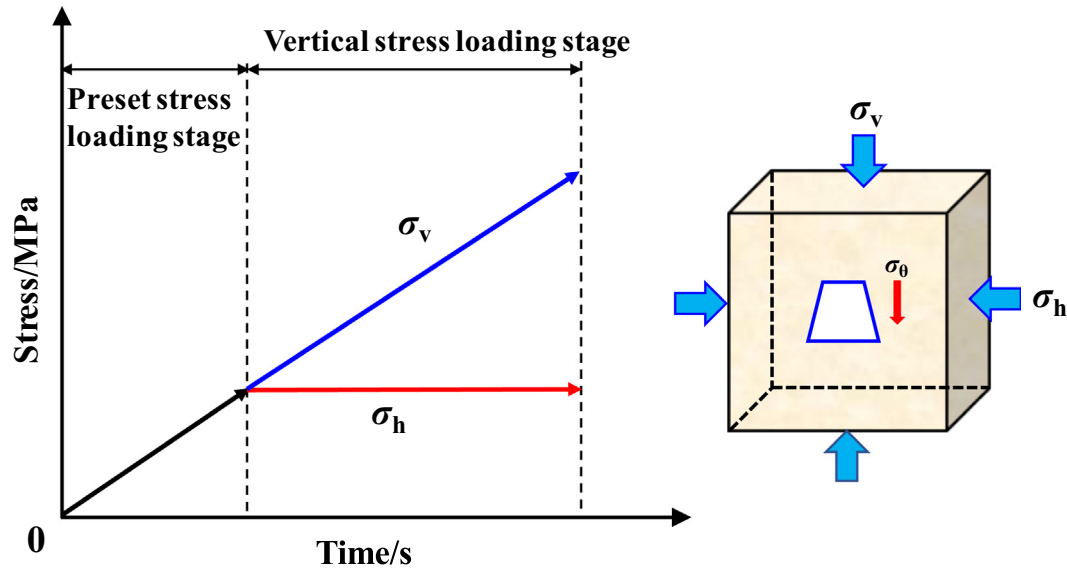


Fig. 3. Schematic illustrations of loading path: (a) schematic diagram of biaxial compression; (b) loading stress path.

3. Experimental results and analysis

3.1. Analysis of rockburst tendency of sandstone

We used the rock brittleness to estimate the rockburst tendency of sandstone. It is defined by an index of the ratio of uniaxial compressive strength to tensile strength of rock, as follow:

$$B = \frac{\sigma_c}{\sigma_T} \quad (1)$$

where, σ_c is the uniaxial compressive (MPa), σ_T is the tensile strength (MPa). The criterion of rockburst tendency proposed by Qiao and Tian [28] shows that:

1. B less than 14.5, then violent rockburst;
2. $B = 14.5\text{--}26.7$, then strong rockburst;
3. $B = 26.7\text{--}40$, then weak rockburst;
4. B greater than 40, then no rockburst.

According to Eq. (1) and experiment result described in Section 2.1, The B value of sandstone is 14.2 and is less than 14.5, indicating the property of violent rockburst.

3.2. Effect of lateral stress on structure model strength and AE evolution

Fig. 4 shows the time history curves of the σ_v , AE energy and cumulative AE energy under different lateral stresses. As it can be seen that the variation of curves under three lateral stresses was significantly different. Under the uniaxial compression condition, i.e., $\sigma_h = 0$ MPa (Fig. 4a), when σ_v increased to about 34.27 MPa ("A" point), the AE energy-time curve increased with small amplitude. It indicates that the cracks begin to initiate within the specimen, which lead to a slight release in the previous cumulative energy. However, the specimen was still in the stage of energy accumulation. Afterward, the specimen entered the stage of accelerated crack extension. With the continued increase σ_v , the cracks gradually propagated to fractures, and the energy input by the experimental machine was transformed into dissipated strain energy. When σ_v reached 47.17 MPa ("B" point), the AE energy and cumulative AE energy both increased sharply. More-

over, a small decrease occurred in the stress-time curve, which was caused by cracks merging into longer microfractures. It might represent the onset of unstable cracks development, which is characterized by significant changes in the structure of the specimen [29]. However, the main fractures did not form at this time, and the supporting capacity of the specimen continued to increase. Thereafter, a considerable amount of AE energy was detected before a drop in axial stress occurred, and the cumulative AE energy began to increase rapidly in a step-like manner from the "A" point. This implied that substantial cracks were generated and then grew, interconnected, and eventually formed main fractures during this period. Finally, the AE energy sharply increased, indicating that a substantial amount of elastic strain energy was released through the fracturing [18]. The values of AE energy and cumulative AE energy of the sandstone occurring at the stress peak point ("C" point) were 5041 mV, and 14809 mV, respectively.

Fig. 4b and c revealed that the σ_v experienced several episodes of sharp reduction accompanied by sharp increases in AE energy, and the cumulative AE energy curve increased in a step-like manner. The maximum AE energy illustrated in Fig. 4b and Fig. 4c, 3859 mV and 6009 mV ("A" point), respectively, occurred at the points of sharp drops in stress rather than at the peak stress point ("B" point). As explained by Taheri et al (2016b), these points might correspond to the crack damage stress point, which were associated with the axial strain nonlinearity, or the reversal point of the volumetric strain at the onset of dilation. Thus, it is reasonable to conclude that the point of sharp decreases in stress correspond to the formation of localized fractures, which were induced by the initiation and propagation of cracks. After σ_v sharply decreased, the supporting capacity of the specimen continually increased, which might be attributed to the structural stability of fractured rock by self-adjustment [24] and the inhibitory effect of lateral stress on fracture growth.

Fig. 5 shows the time history curves of cumulative AE energy under different lateral stresses. When σ_h was not applied or was maintained constant at 10 MPa, the curve of the cumulative AE energy was relatively flat at the initial stage of stress loading. This meant that only a negligible amount of energy was released during this period, which was mainly manifested as the process of energy accumulation [30]. However, when σ_h reached 20 MPa, the elastic strain energy changed from accumulation to released states at the

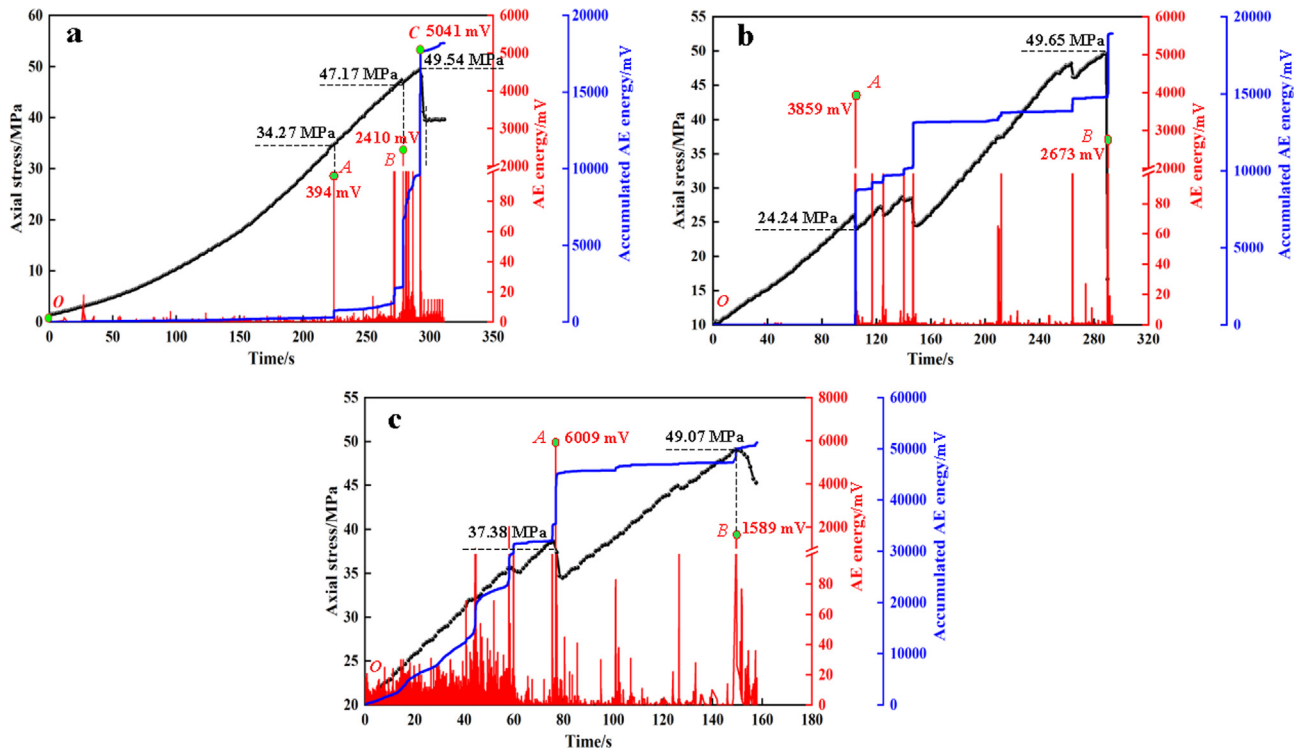


Fig. 4. Stress-time curves associated with AE energy and accumulated AE energy under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa.

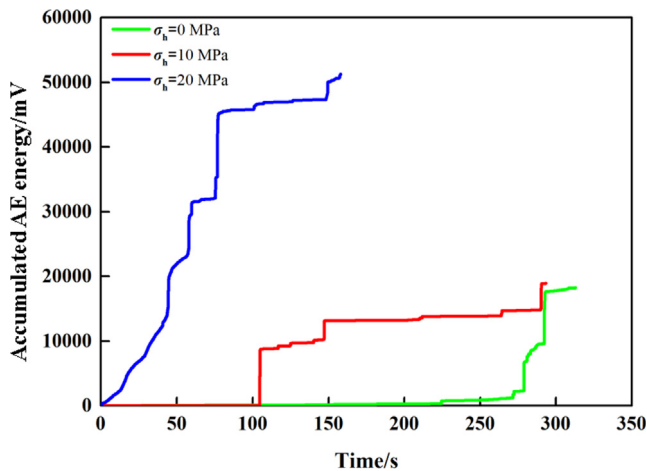


Fig. 5. Time history curves of the cumulative AE energy under different lateral stresses.

initial stage of stress loading, and the time required for failure was shortened. This means that a high lateral stress has a significant influence on the rate of the development from crack initiation to macroscopic fracturing.

3.3. Strain distribution around the opening under different lateral stresses

The initiation and extension of cracks can cause the rock grains to dislocate, which increases the distance between grains and inevitably causes the localized strain. Dzik and Lajtai [31] found that the strain gauge in the vicinity of the nucleation point can show some signs of disturbance. Thus, in our biaxial experiments, the measured strain gauges data could be used to illustrate the fracture process of rock around the trapezoidal opening.

3.3.1. Tangential strain distribution characteristics around the opening under different lateral stresses

The tangential strain distribution around the opening can directly reflect the stress state in rock and distinguishes the fracture type. The curves of tangential strain-time under different lateral stresses were obtained by using the strain gauges, which were fixed around the opening, as shown in Fig. 6. A positive value of the strain corresponds to tensile strain, whereas compressive strain is denoted as a negative value. Furthermore, according to the experimental results demonstrated by Carter et al. [32], the deflection of the load-strain curve, which was replaced by the time-strain curve in the present study, can reflect the process of fracture. A rapid increase in the extensional strain denotes that a fracture crosses beneath the strain gauge. Conversely, strain relief is caused by the fracture approaching and then bypassing the strain gauge. From Fig. 6, it can be seen that when $\sigma_h = 0$ or 10 MPa, the tensile strains in the shallow surrounding rock at the locations of roof and floor were both firstly increased and then decreased. The deflection points of the time versus tangential strain curves at strain gauges R3 and F7 indicated the primary fracture nucleation, and the inflection points of the time-strain curves at strain gauges R3 and F7 implied that the tensile fracture formed in the inner surface of the opening and then bypassed the strain gauge, which led to strain relief [23]. For the deep surrounding rock, such as 5 mm and 15 mm from the opening, the roof and floor were basically in a slightly compressive strain state, and the strains changed greatly only when the specimen was about to be instability. Compared with aforementioned fracture process under relatively small lateral stress, that when $\sigma_h = 20$ MPa, there were several sudden drops of tangential tensile strain in the locations of roof and floor (strain gauges R3 and F7), which indicates the formation of the localized damage. However, the time versus tangential strain curves representing strain gauges R3 and F7 showed an upward trend. This may be due to the fact that the cracks formed near the strain gauges R3 and F7 were restrained under the action of high lateral stress, and caused the strain gauges R3 and F7 to be

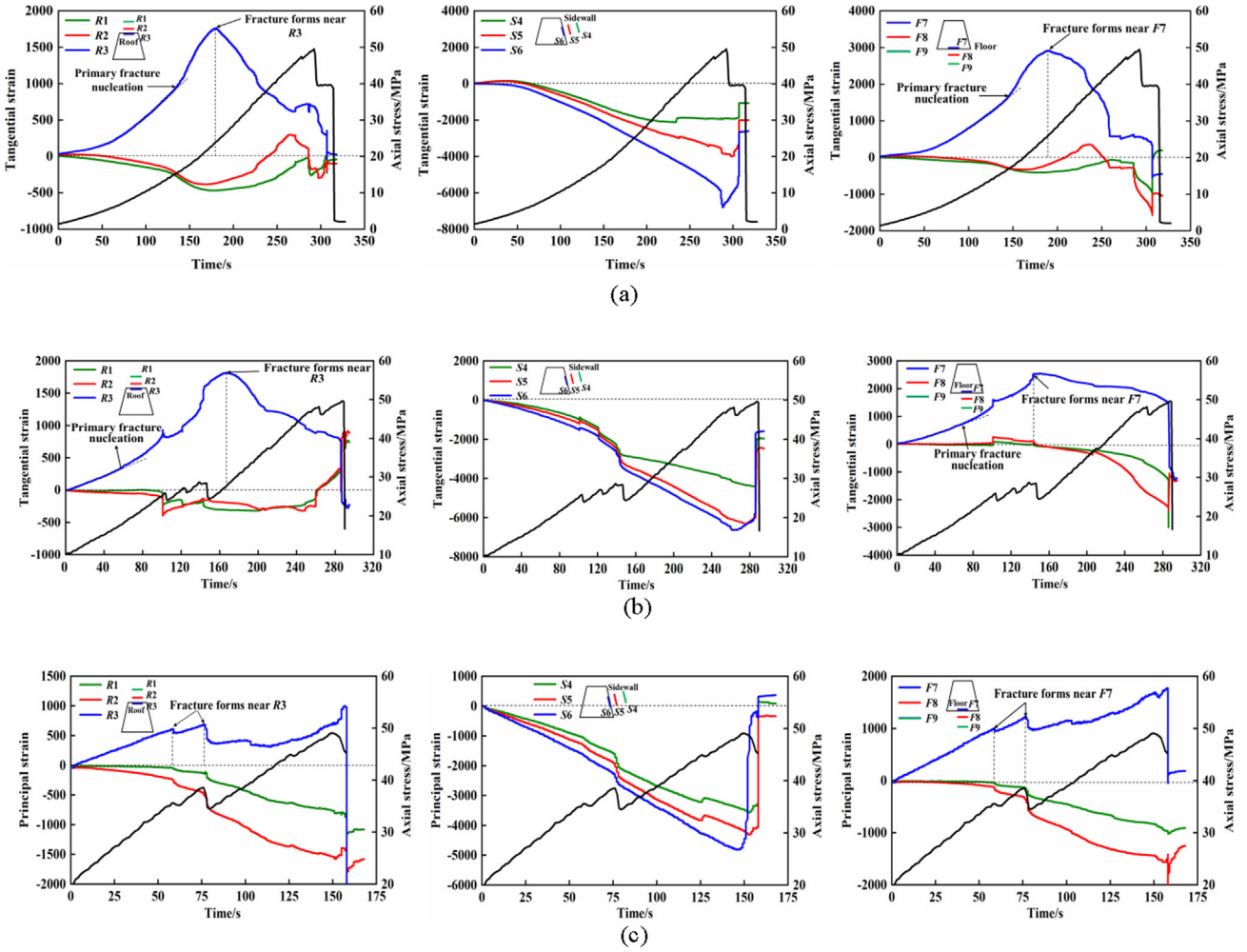


Fig. 6. Tangential strain response to biaxial compression in the locations of roof, sidewall and floor under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa.

still affected by the tangential tensile stress. For the deep surrounding rock, such as that 5 mm and 15 mm from the opening, the tangential compression strains gradually decreased, and its magnitude was greater than that under the conditions of $\sigma_h = 0$ or 10 MPa. It indicates that at the locations of roof and floor of trapezoidal opening, the influence regions of compressive stress obviously increase when the lateral stress increases to a critical value, and it is necessary to strengthen the support in deeper regions under this level of lateral stress.

For the sidewall of trapezoidal opening, the tangential compressive strains increased linearly under different lateral stresses and gradually decreased with an increase in the distance from the opening. Note that the tangential compressive strains under the conditions of $\sigma_h = 10$ and 20 MPa increased in steps, and each rapidly increase of the tangential compressive strain corresponded to the rapidly decrease of the σ_v , which indicated that local fracture or even spalling was formed at this position [33]. Otherwise, the magnitude of the tangential compressive strains at strain gauges S4 and S5 was much larger than that at strain gauges R1, R2, F8, and F9. It indicates that the tangential compressive stress in the deep surrounding rock of the sidewall is greater than that in the deep surrounding rock of the roof and floor, which can cause the deep surrounding rock of the sidewall to be damaged more severely.

In accordance with the above-mentioned analysis, the surrounding rock at the boundary of roof and floor was subjected to

tensile stresses, and the compressive stress dominated the failure of sidewall. Thus, it can be deduced from the tangential strain distribution around the opening that tensile fractures would be formed in the roof and floor, and spalling failure would occur as a result of a large number of cracks initiated and propagated at the sidewall that were induced by the compressive stress [19].

3.3.2. Principal strain characteristics in typical locations under different lateral stresses

Two 3D strain rosette gauges (C10 and C11) were fixed in the corner near the opening, shown in Fig. 1c. The principal strains ε_1 and ε_3 and the principal stresses σ_1 and σ_3 were calculated on the basis of the obtained strains ε_x , ε_y and ε_u , respectively (Fig. 7), where the negative value denotes compression, and the tensile strain and stress are represented as positive values. The ε_1 and ε_3 directions are perpendicular to each other. The principal direction ψ is the angle of ε_1 to the horizontal direction, in which counter-clockwise rotation is positive and clockwise rotation is negative [24]. ε_1 , ε_3 , σ_1 , σ_3 , and ψ can be expressed as follows:

$$\begin{aligned}\varepsilon_{1,3} &= \frac{\varepsilon_x + \varepsilon_y}{2} \pm \frac{1}{2} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + (2\varepsilon_u - \varepsilon_x - \varepsilon_y)^2} \\ \sigma_{1,3} &= \frac{E}{2} \left[\frac{\varepsilon_x + \varepsilon_y}{1 - \mu} \pm \frac{1}{1 + \mu} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + (2\varepsilon_u - \varepsilon_x - \varepsilon_y)^2} \right] \\ \psi &= \frac{1}{2} \tan^{-1} \left[\frac{2\varepsilon_u - \varepsilon_x - \varepsilon_y}{\varepsilon_x - \varepsilon_y} \right]\end{aligned}\quad (2)$$

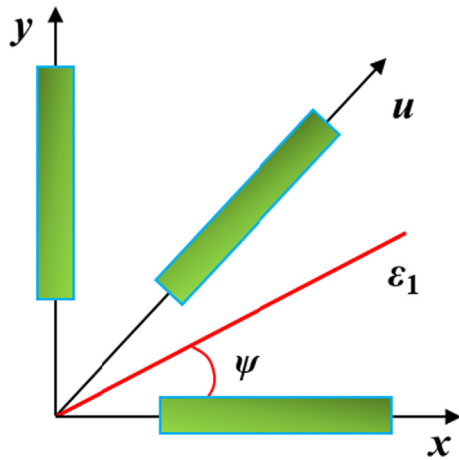


Fig. 7. Schematic diagram of three-dimensional strain rosette gauge.

where E is Young's modulus of sandstone (GPa), and u is the Poisson's ratio.

Fig. 8 shows the variation in principal strain ε_1 and ε_3 and principal direction ψ , where ε_1 was a positive value and represented the principal tensile strain, and ε_3 was a negative value and represented the principal compressive strain. As shown in Fig. 8, the principal compressive strains increased proximately in the linear direction. However, the principal tensile strains increased slowly or were approximate constant with the increase in the vertical load. In the aspect of the principal strain magnitude, principal values of compressive strain at the upper and lower corners were both much larger than principal tensile strain under different lateral stresses. It may imply that compressive fractures would be preferentially formed at the corners of trapezoidal opening. In addition, according to Eq. (2), we calculated the principal stresses, and found that under different lateral stresses, the compressive stresses at the lower corners were both greater than those at the upper corners (Fig. 9), which implied that the compressive stress concentration at the lower corner was more intense than that at the upper corner. This result might indicate that compared with the rock at the upper corner, the rock at the lower corner would enter the stage of cracks initiation and propagation earlier.

In the aspect of principal strain orientation, the experimental results of Wu and Pollard [34] showed that secondary fractures can be initiated and propagated from a set of primary fractured in a new direction parallel to the maximum principal stress and that the stress field orientation controls the development of the secondary fractures. As seen in Fig. 8, with the increase of the vertical load, the ψ rotated clockwise from the initial value to close to 0° or less than 0° under the conditions of $\sigma_h = 0$ and 10 MPa. Meanwhile, the cracks propagated along the direction of the principal compressive stress. However, when $\sigma_h = 20$ MPa, the ψ barely rotated and kept in about 0° , which meant that the principal compressive strain ε_3 was approximately parallel to the vertical load direction. Hence, it is reasonable to confirm that with the rotation of the principal stresses, new microcracks are initiated over a series of critical angles, and lead to the crack density and the extent of stress-induced brittle fracture damage of the rock at the corner under the conditions of $\sigma_h = 0$ and 10 MPa are greater than that under the conditions of $\sigma_h = 20$ MPa [35]. In practical engineering, with the propagation of microcracks along different directions within the surrounding rock near the corners and the interaction of localized stress (strain) concentrations around the tips of the cracks, the connectivity of microcracks is increased, and complex seepage channels are then formed. This significantly increases

the permeability of the surrounding rock near the surface of opening and leads to coal and gas outbursts or water inrush.

On the basis of above experimental results, we can infer that the corners are easy to form compressive fracture zones under the combined action of high concentrated stress and the rotation of principal stress, and a higher lateral stress can restrain the rotation of principal stress and consequently reduce the crack density in the corners. Besides, the damage degree at the lower corner was greater than that at the upper corner. Thus, the support at the corner should be strengthened during the excavation of tunnels or underground roadways, particularly strengthen the support at the lower corner. Due to the boundary with a circular arc can alleviate the stress concentration [24], it is also suggested that the localized rounding of the corner should be applied to tunnels and underground roadways to avoid premature failure caused by the excessive stress concentration.

4. Analysis of rockburst process under different lateral stresses

4.1. Fracture modes under different lateral stresses

The initiation and propagation of three basic cracks, including tensile cracks, shear cracks, and a combination of the two, are generally accompanied by the failure process of rock material. Since the incidents of crack initiation and propagation have different AE signatures depending on the cracking modes, these differences can be applied to distinguish the fracture mode [36]. Based on the above AE characteristics, many scholars have used the ratio of the average frequency value (AF) to the rise angle value (RA) to identify different fracture modes, particularly in concrete material [36–39]. However, application of this ratio to rock material is rare.

In order to distinguish the crack types, the Brazilian split tests and direct shear tests were conducted to obtain the AF/RA characteristics of the tensile and shear cracks. This was done because the tensile cracks are dominant in Brazilian split tests, while shear cracks accounted for most during direct shear tests [18]. According to the results of these two tests, the value of AF/RA was defined as 60 to differentiate the mode of source cracking as shown in Fig. 10. If the ratio of AF/RA is greater than 60, the AE signal originates mainly from tensile cracks; otherwise, the AE signal comes from shear cracks. Fig. 11 shows scatterplots of AF and RA under different lateral stresses. We can find that the AE signals representing both tensile cracks and shear cracks appeared under different lateral stresses, and the distribution of AE signals representing tensile crack was basically consistent. The main differences between them were concentrated in the distribution of AE signals representing shear cracks. When $\sigma_h = 0$ MPa, only a relatively small number of shear signals appeared during the entire process of specimen failure. It indicated that with the increase of σ_h , the shear cracks gradually increased and the failure mode of the specimen shifted from tensile fracture to mixed tensile and shear fracture. This result is related to the application of lateral stress constraints. Tensile fracture typically occurs in unconfined compression tests [40].

After specimen failure, the specimens were scanned by applying medical CT scanning to more effectively examine the interior fracture morphologies. Furthermore, the 3D fracture network of specimens subjected to different lateral stresses was reconstructed depending on the acquired CT images. In order to observe the interior fracture morphologies clearly, we selected the side views of the reconstructed specimens, as shown in Fig. 12. The specimens showed significant variations in the failure mode under different lateral stresses. It was evident that penetrating macro-tensile fractures and inclined macro-shear fractures formed along the axial stress direction when the lateral stress was loaded ($\sigma_h = 10$ and 20 MPa), and the number of shear fractures increased with the

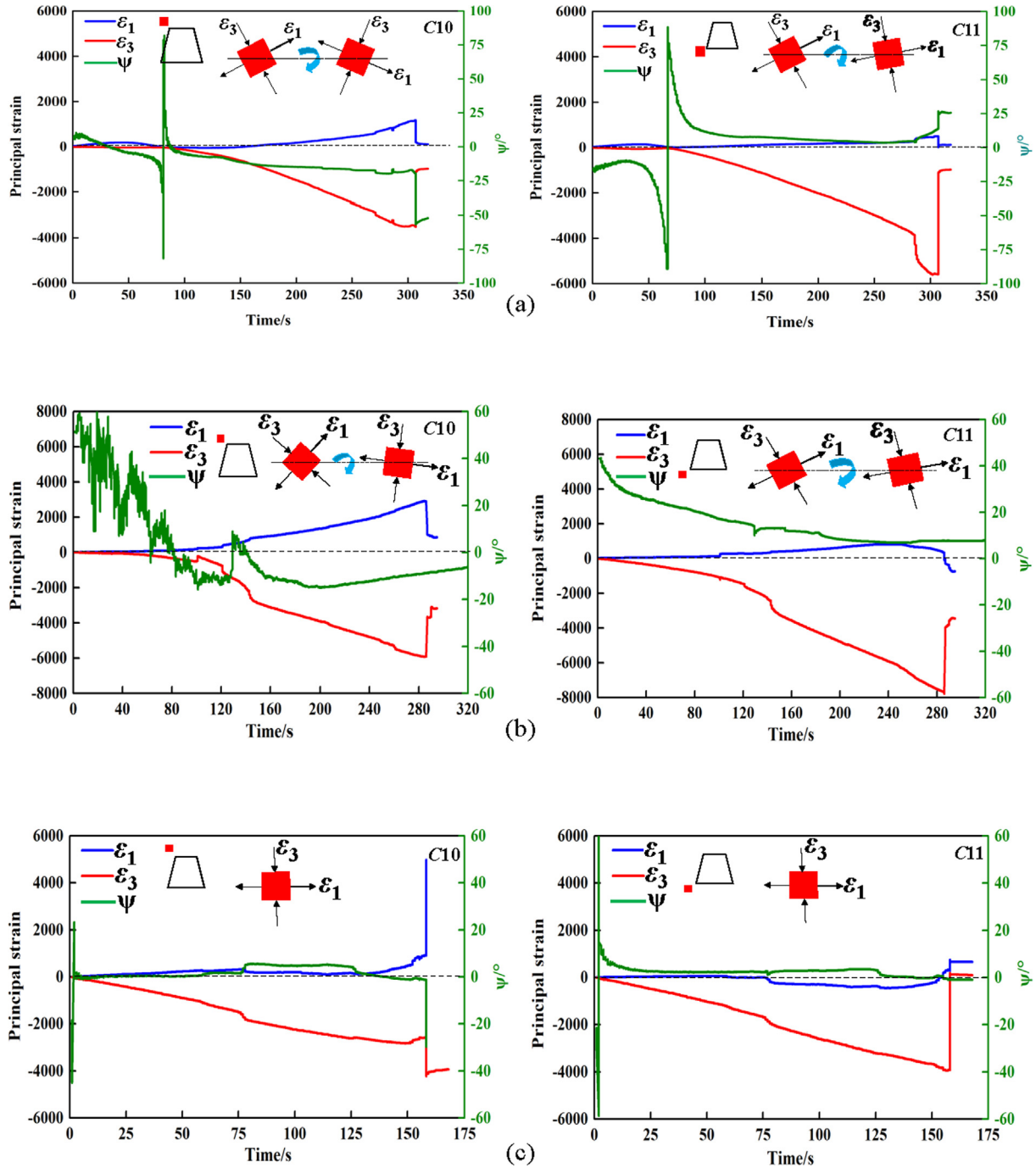


Fig. 8. Principal strains and direction in two typical locations under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa. C10 and C11 denote the locations of the upper corner and lower corner, respectively.

increase of σ_h . When $\sigma_h = 20$ MPa, a set of X-shaped inclined shear fractures was observed inside the specimen. However, when $\sigma_h = 0$ MPa, the penetrating shear fractures along the axial stress direction were replaced by the localized shear fractures, and the instability of the specimen was mainly caused by the localized tensile fractures and localized shear fractures. The above results were consistent with the variation of AE signals reflecting the shear cracks in Fig. 11. In addition, as shown in Fig. 12, the number of fractures passing through the opening gradually increased with the increase of σ_h , which indicates that the damage degree of underground roadways is positively correlated with the lateral stress.

4.2. Characteristics analysis during the rockburst process

Typical failure phenomena recorded by using high-speed camera are shown in Fig. 13. Because the recording time of the high-speed camera used in our experiments is 4 s, the pre-failure process of the opening was difficult to obtain. Thus, at the beginning, clear spalling was observed in the region of the left sidewall. The mechanism of this phenomenon was caused by the propagation of extensile fractures subparallel to the maximum compressive stress direction or opening boundary [18]. As the vertical loading continued to increase, the unstable cracks grew to the point of creating rock flakes of buckling size, and then we observed the

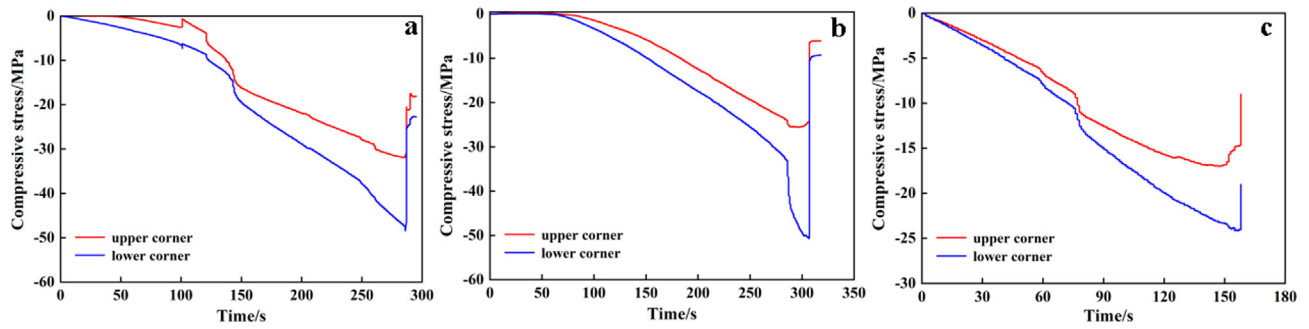


Fig. 9. Compressive strains measured at the corners under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa.

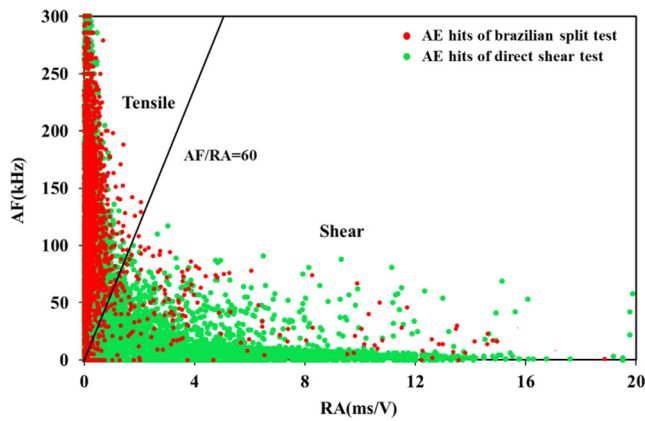


Fig. 10. Scatter plot of AF and RA.

separation of rock flakes from the surface of opening. Afterward, the residual strain energy after cracking or fracturing began to be released in the form of kinetic energy, which was manifested by a rapid ejection of rock particles and fragments into the opening space accompanied by loud sound and the breaking of flakes under the intense impact. When $\sigma_h = 0$ MPa, we found that the intensity and number of particles ejection during the rockburst were higher than those under the lateral stress of 10 or 20 MPa, which indicates that under uniaxial compression condition, the kinetic energy released at the time of structural model instability was larger. The main reason for this phenomenon is that the loading of lateral stress results in the existence of high stress concentration around the corners of the trapezoidal opening [26], and this stress concentration increases with the increase of the lateral pressure coefficient [20]. Therefore, under the biaxial compression condition, the cracks and localized damage can be formed earlier around the corners before the specimen became completely unstable, which led to the release of the previous cumulative elastic strain

energy. As a result, the kinetic energy converted from the residual elastic strain energy at the time of structural model instability became less. In this term, the initiation and propagation of cracks can consume a large amount of elastic strain energy. Hence, it is advantageous to hold some cracks in the rock around the opening to release the cumulative elastic strain energy, and then weaken the intensity of rockburst. In addition, during the rockburst process, the ejection of particles always occurred at the lower corner of the trapezoidal opening to form a distinct fragmentation zone. However, this phenomenon was not observed to a significant degree at the upper corner, which caused by a higher concentrated compressive stress at the lower corner, compared with that on the upper corner, as discussed in Section 3.3.2. It suggested that we need to strengthen the support near the lower corner of tunnel or underground roadway.

4.3. Failure characteristics of sidewall

Fig. 14a-c shows that the V-shaped notches were formed on the sidewall along the opening axis after the experiment, which was similar to the typical characteristic at the engineering site (Fig. 14d). In order to reflect the change of model failure mode caused by rock burst around the chamber under different lateral stress, the damage depth and angle of the V-shaped notch were accurately measured, as shown in Fig. 15. From Fig. 15, we can see that the damage depth increased and the damage angle decreases with the increase of the lateral stress. It indicates that the damage degree at the sidewall of the trapezoidal opening increases with the increase of the lateral stress when the rockburst occurs, which may be due to the high stress concentration around the opening caused by the high lateral stress. Note that the width of the V-shaped notch under different lateral stresses is approximately equal to that of the sidewall, i.e., the damage was distributed on the whole sidewall of trapezoidal opening, rather than in the middle of sidewall like circular opening [19]. To verify this conclusion, the biaxial compression experiments were carried

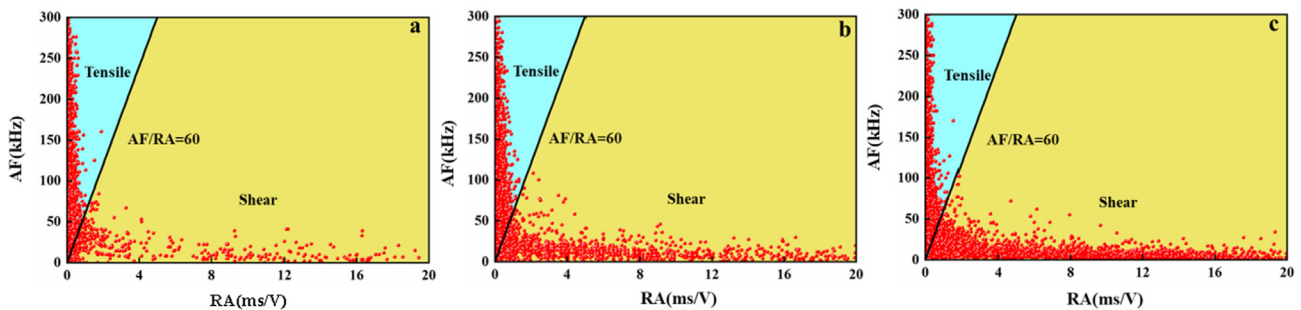


Fig. 11. Scatter plots of AF and RA under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa.

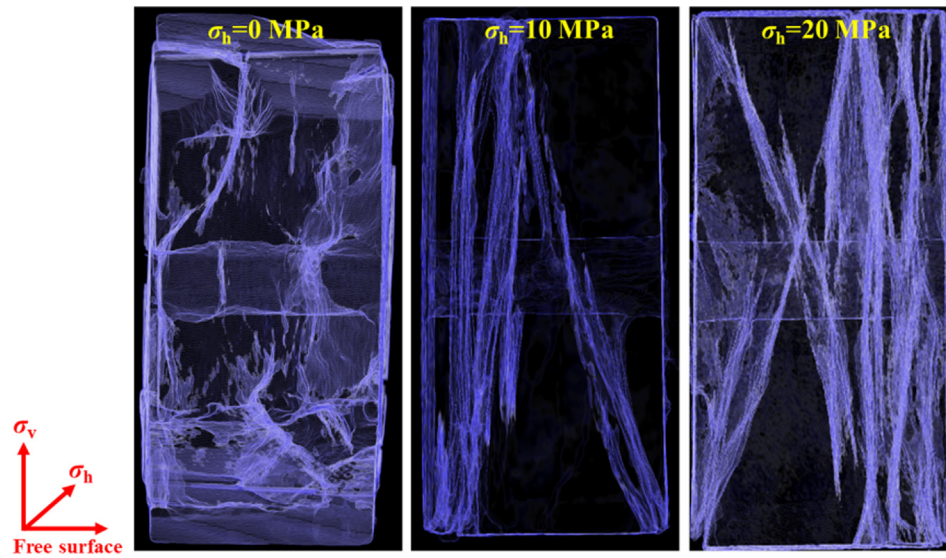


Fig.12. Binary cross-section of inner fracture of specimens observed after failure under different lateral stresses.

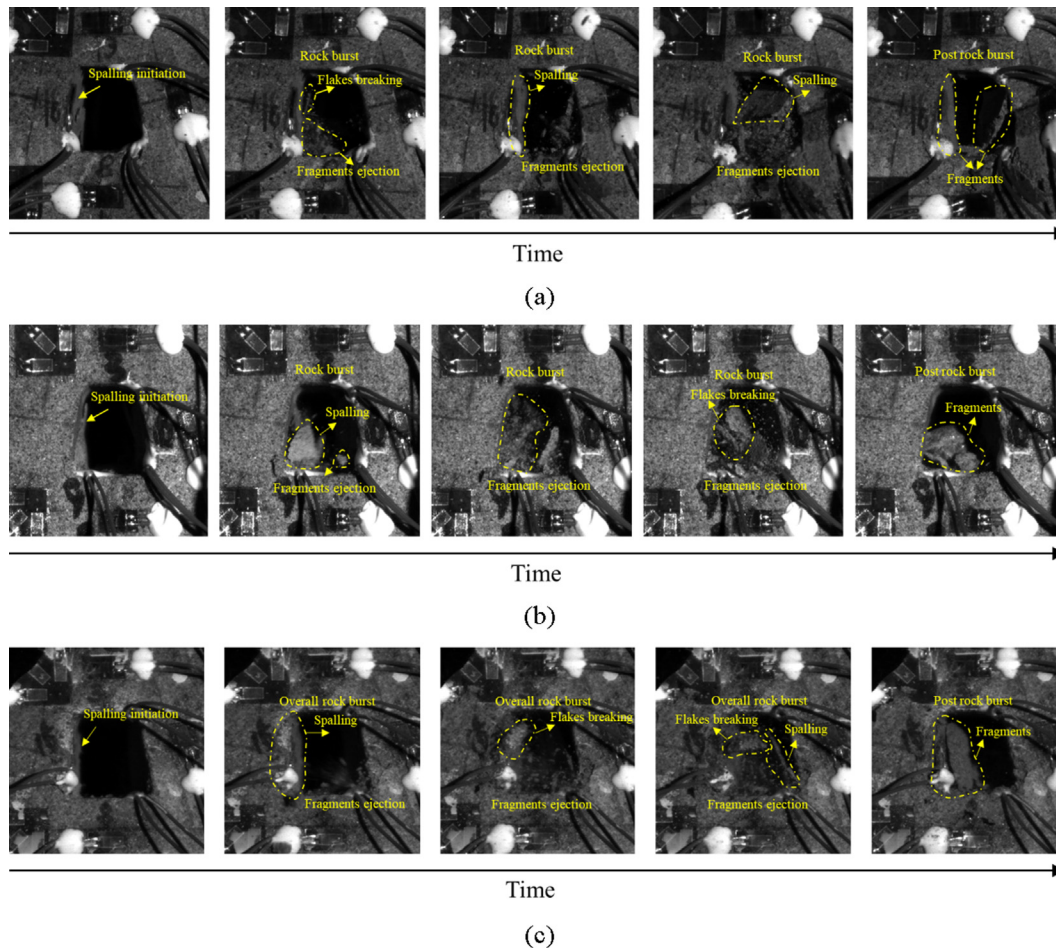


Fig. 13. Typical failure phenomena recorded by using high-speed camera under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa.

out on the circular opening with a diameter of 15 mm under the same experimental conditions (i.e., the lateral stresses are 0, 10 and 20 MPa). The CT images of trapezoidal opening and circular opening are shown in Fig. 15. We can observe that the whole sidewalls of trapezoidal opening are damaged to a certain extent, while

the damage area of circular opening is mainly concentrated at the middle part of sidewall. The damage area at the sidewall of trapezoidal opening is significantly larger than that at the sidewall of circular opening, which implies that the supporting area of trapezoidal opening should be larger than that of circular opening. This

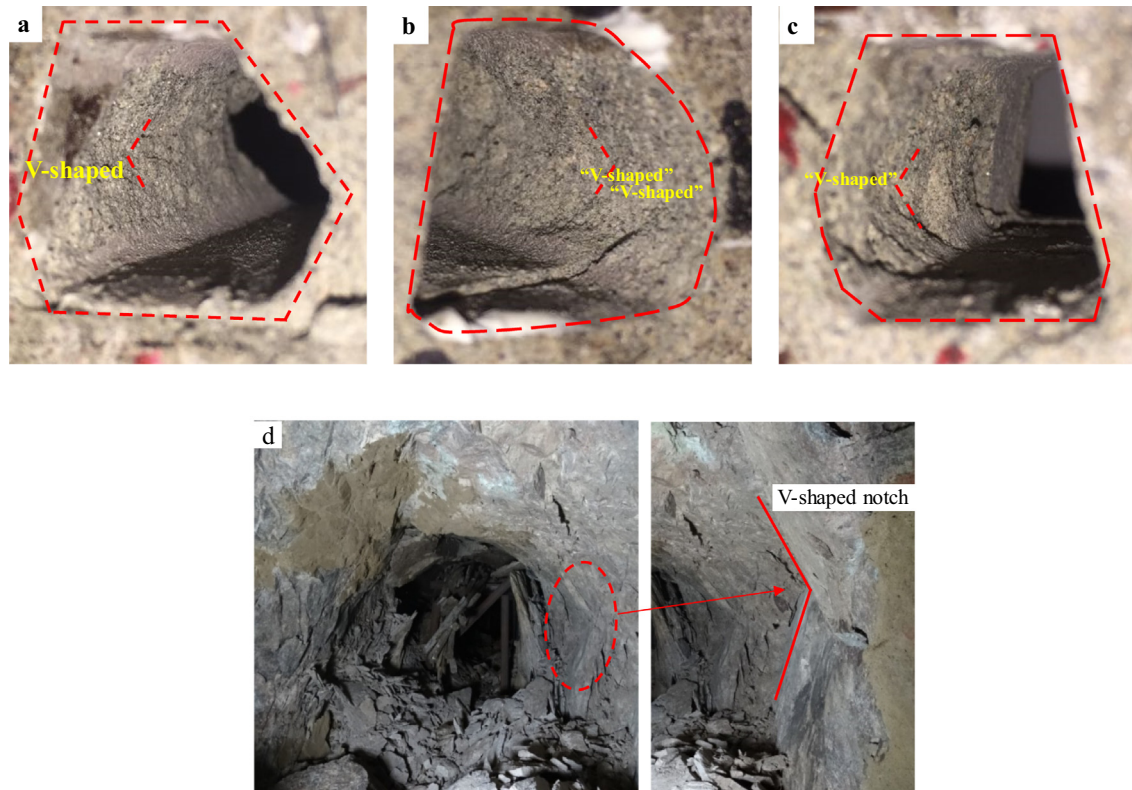


Fig. 14. V-shaped failure bands after rockburst under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa; (d) rockburst in tunnel at a gold mine site [41].

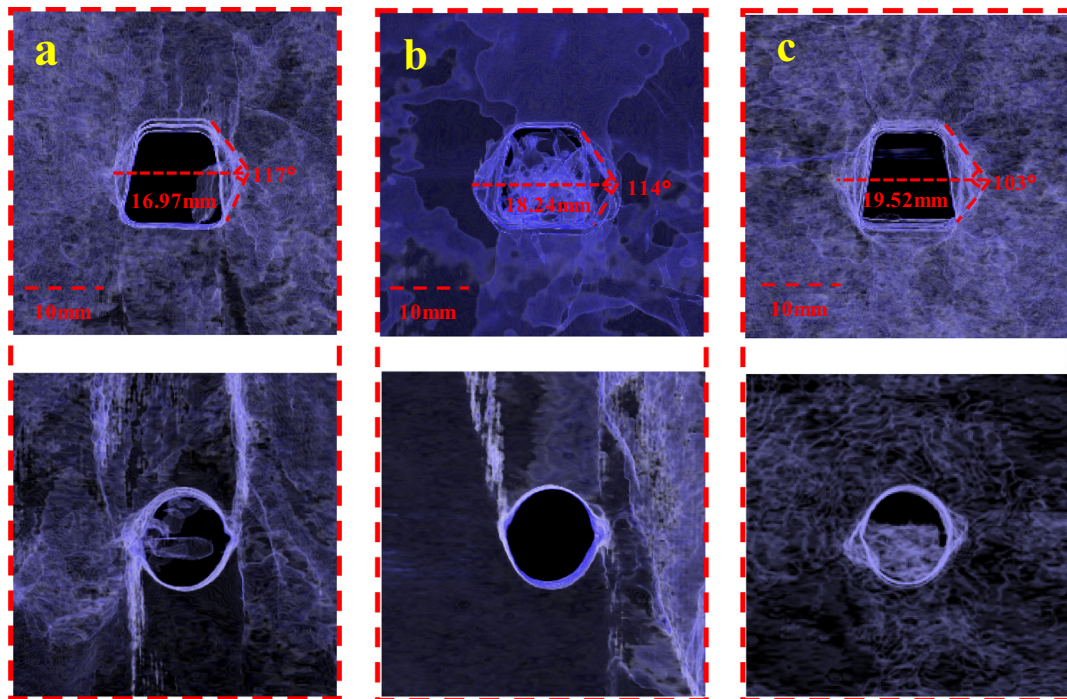


Fig. 15. The CT images under different lateral stresses: (a) $\sigma_h = 0$ MPa; (b) $\sigma_h = 10$ MPa; (c) $\sigma_h = 20$ MPa.

is due to that the corner of trapezoid opening is a common stress concentration area, which is easy to cause the initiation and propagation of cracks. As the tangential stress increases, the cracks at the corner converges with that at the sidewall, a thin rock slab with a width close to that of sidewall is formed.

The failure process on the sidewall of trapezoidal opening could be summarized in Fig. 16. As the tangential compressive stress increases and concentrates in the sidewall and corner, wing cracks initiated and propagated (Fig. 16a). When these wing cracks are connected with each other (Fig. 16b), thin rock slabs with width

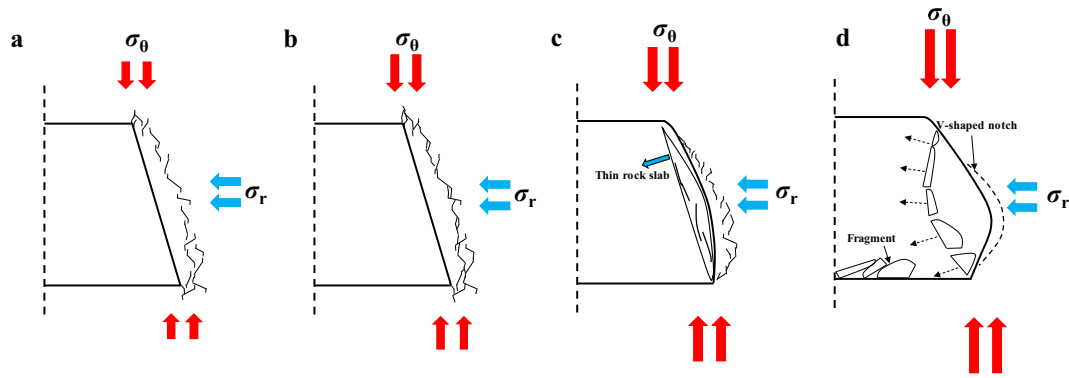


Fig. 16. Schematic illustration of rock fracture near the free surface of opening: (a) crack initiation and propagation; (b) crack coalescence; (c) spalling failure; (d) formation of V-shaped notch. σ_θ and σ_r are tangential stress and radial stress, respectively.

approximately equal to that of the sidewall are formed on the surface of the sidewall (Fig. 16c). As the tangential compressive stress increases, these thin rock slabs begin to bend and store strain energy until it buckles. Subsequently, these thin slabs break off and are ejected into the space of opening (Fig. 16d). This process will establish a new free surface, and the process will repeat itself until the surrounding rock around the opening is in complete failure. In addition, crack initiation and propagation are responsive to the radial stress, and the increase of radial stress can lead the crack propagation to cease [42]. Besides, the stress gradient exists around the opening [43]. The radial stress increases and the tangential stress decreases with an increase in distance from the opening boundary. As a result, the ratio of radial stress/tangential stress gradually increases, and it is negatively correlated with the fracture length [44]. Therefore, inner fractures with smaller dimensions will appear as a result of a larger ratio of radial stress/tangential stress. In summary, the V-shaped notches with width approximately equal to that of the sidewall were formed after rockburst occurs (Fig. 16d).

Fig. 17 shows that the rock powders and slabs are produced during the failure process of the sidewall. The rock slabs formed in the sidewall exhibit a thin morphological feature in both wings and thick in the middle, which is consistent with the characteristics of spalling failure in the sidewall obtained by many scholars [15,19,25,45]. In particular, it should be noted that rock slabs formed by spalling failure in the sidewall have been produced before the specimen failure, as shown in Fig. 13. These rock slabs did not completely separate from the sidewall until the specimen failure. In addition, when the specimen was completely destroyed, it mainly produced the macroscopic fractures surface approximately parallel to the direction of vertical load (Fig. 12), and released residual elastic strain energy to separate the rock slabs from the sidewall. Thus, the formation of V-shaped notches is mainly dominated by the spalling failure in the sidewall. Fig. 18-

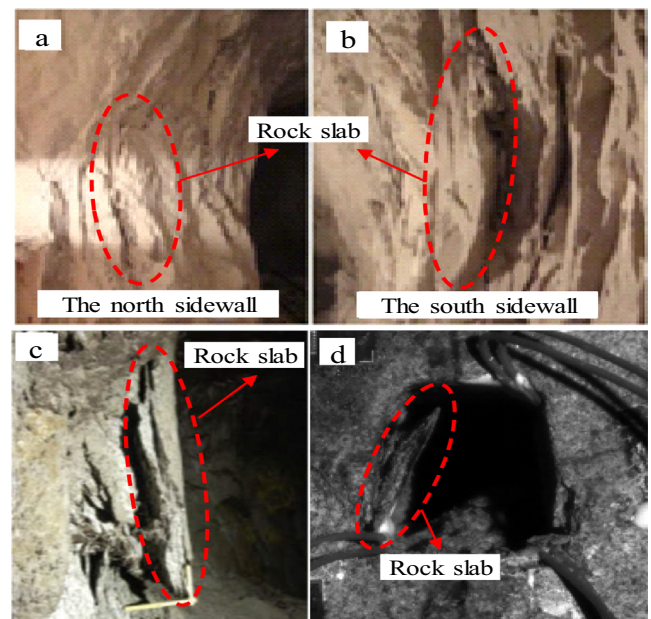


Fig. 18. Spalling failure in the field tunnels and experiment: (a) and (b) thin slabs formed by spalling failure in the test tunnel of Jinping II hydropower station [46]; (c) surface spalling failure at the interface between the stress-fractured and undisturbed rock mass [9]; (d) spalling failure in the experiment.

a-d presents the typical spalling at the sidewall both on-site and in the experiment. Surface spalling occurred at the sidewall, forming a large piece of thin rock slab (Fig. 18a-c), which is similar to that of rock slab at the sidewall in Fig. 18d. This indicates that the failure mode of the opening sidewall is consistent with that of the field tunnels, and both exhibit typical spalling features. In

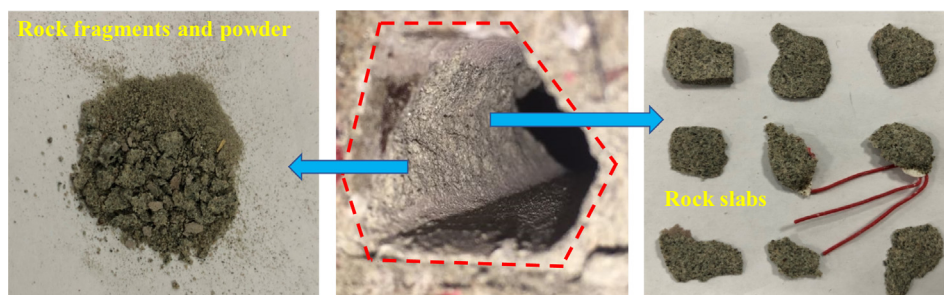


Fig. 17. Rock powders and rock fragments formed during the experiment.

summary, the experiment in our study provides a relatively realistic simulation of the failure process of the tunnel.

4.4. Discussion of engineering guidance

As a common shape of tunnel section, trapezoidal opening has a smaller exposed area of the roof than rectangular opening and it is easy to construct, and able to bear certain lateral stress [47]. According to the experimental results, the high lateral stress accelerates the release of elastic strain energy and the initiation and propagation of cracks can consume a large amount of elastic strain energy. Thus, on the basis of ensuring the stability of the tunnel, we can artificially create cracks in the surrounding rock to consume the accumulated elastic strain energy (e.g., hydraulic fracturing and hydraulic slotting). In addition, the high lateral stress can restrain the stress rotation at the corner of trapezoidal opening, i.e., restrain the increase of fracture density and the degree of brittle fracture damage [35]. Conversely, the high lateral stress causes a wider damage at the sidewall of the trapezoidal opening. Therefore, when selecting the excavation direction, we need to comprehensively consider the magnitude of each horizontal stress. When the difference between the two horizontal stress is slight, the excavation of the trapezoidal tunnel perpendicular to the maximum horizontal stress is conducive to restrain the stress rotation at the corner. When the difference between the two horizontal stress is large, the excavation along the maximum horizontal stress is conducive to control the damage degree at the sidewall. In order to prevent the integral spalling failure at the sidewall in the case of rockburst, the full profile support by rock bolting-injection, mesh and shotcrete surface support should be applied so as to form a composite support structure, especially strengthen the support around the corner.

5. Conclusions

In this study, sandstone specimens containing trapezoidal openings were used to conduct biaxial compression experiments. The effect of lateral stress on the AE characteristics, stress distribution, rockburst process, and failure mode were systematically discussed. The main conclusions are given in the following points.

- (1) A high lateral stress accelerates the formation of localized fractures during the loading process accompanied by the release of elastic strain energy, and the formation of fractures can effectively consume the elastic strain energy. With the increase of the lateral stress, the variation of elastic strain energy changes from accumulation to release state at the initial stage of stress loading and the time required for failure was shortened. This means that a high lateral stress has a significant influence on the rate of the development from crack initiation to macroscopic fracturing.
- (2) The shallow surrounding rock at roof and floor of the opening was affected by tensile stress. While, the deep surrounding rock at the roof and floor were basically in a compressive stress state, and the influence regions of compressive stress obviously increase when the lateral stress increases to a critical value. In addition, the tangential compressive stress at the deep surrounding rock of the sidewall is greater than that at the deep surrounding rock of the roof and floor, which caused the deep surrounding rock of the sidewall to be damaged more severely due to a high compressive stress.
- (3) The principal compressive stress at the lower corner was greater than that at the upper corner. A high lateral stress can restrain the rotation of principal stress and consequently reduces the crack density near the corner.

- (4) With the increase of the lateral stress, the shear signals obviously increased, implying that the failure of the specimen gradually shifted from tensile fracture to mixed tensile and shear fracture. According to the results of CT scanning, the shear fracture zones increased with the increase of the lateral stress.
- (5) The damage degree at the sidewall of the trapezoidal opening increases with the increase of the lateral stress when the rockburst occurs, and the failure zones of trapezoidal opening are larger than those of circular opening, which indicates that the supporting areas on the sidewall of trapezoidal opening need to be larger than those on the sidewall of circular opening.

Ethics statement

It is not relevant to my work.

Data accessibility

Data available within the article or its supplementary materials.

CRediT authorship contribution statement

Honggang Zhao: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Data curation, Funding acquisition. **Chao Liu:** Resources, Writing - review & editing, Supervision, Funding acquisition. **Gun Huang:** Project administration, Visualization, Funding acquisition. **Beichen Yu:** Investigation, Funding acquisition. **Yubing Liu:** Software, Funding acquisition. **Zhenlong Song:** Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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